Cafe Stock Prediction with Machine Learning

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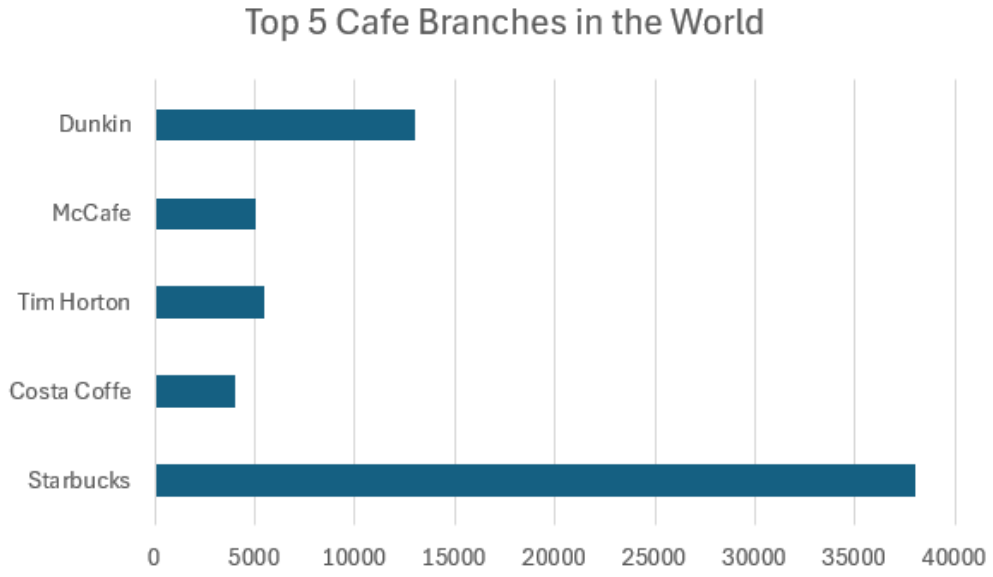
Problem Definition

- As the UN has 17 goals to deal with global challenges that are poverty, pollution, resource problems, inequity, economy. Our design project focuses on those 17 goals. Sustainability is the main purpose of this project by reducing waste of resources.
- Hunger, pollution, waste of resources creates challenges to the globe.
- There are more than 38.000 cafe only for Starbucks around the world. When we consider other organizations, the number is enormous.
- In those cafes there are enormous waste of resources everyday caused by the non-planned resource usage.



Problem Definition

- In most of cafe organizations, there are some materials like milk, bread, cheese, vegetables, fruits that gets non-consumable by the time. This situation creates waste and loss for the company, waste of resources in the world as we consider how many cafes there are around the world. Henceforth, the impact of the saving contributes grand to the sustainability.

Sustainable Solution

- Machine learning prediction algorithm can be implemented for this problem as it makes prediction based on the data by recommending needed amount of product for each day and acting based on the selling of the day and stock in the café.
- Therefore, café makes saving of the materials for each day by avoiding decay of the materials. It leads to sustainability in organizations like cafes.
- Considering the amount of cafe in the world is large, its impact is enormous.

Design

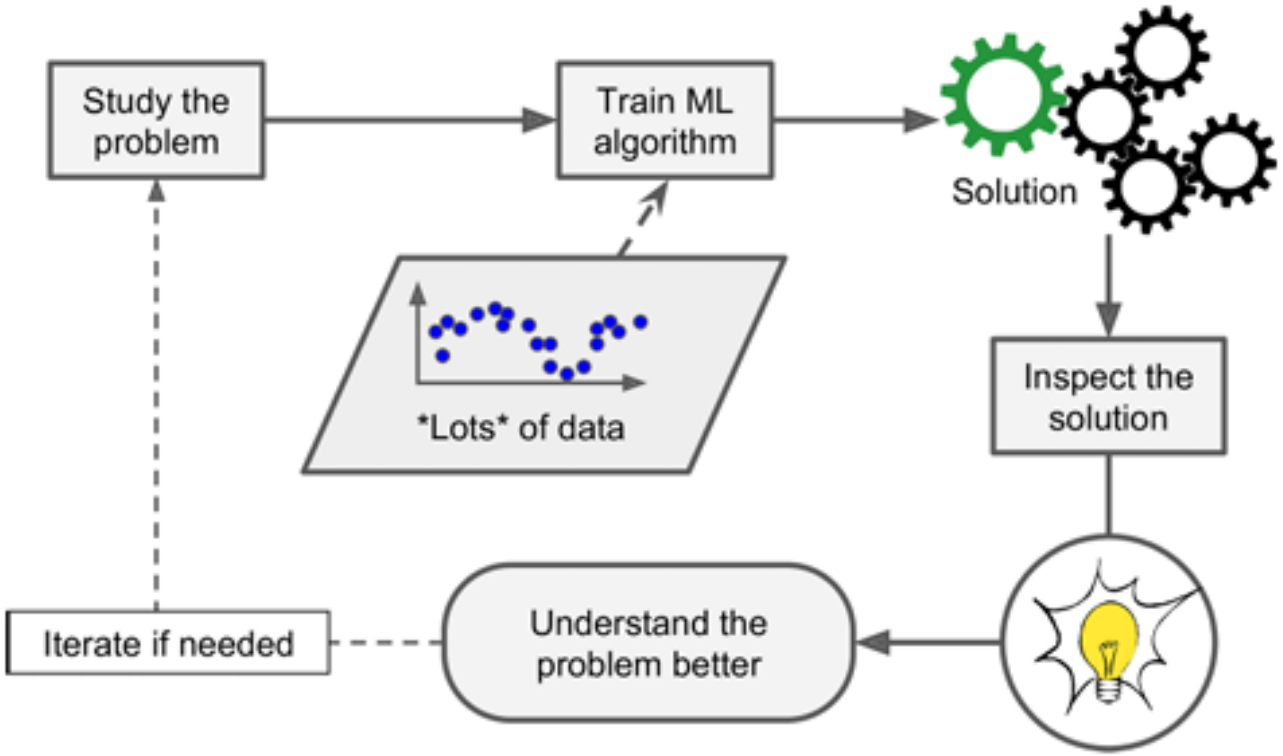


Figure 1-4. Machine Learning can help humans learn

Design

ML algorithm can be implemented in a continuous way fed by previous and present data in a dynamic way.

An application to track the recommendations of the ML algorithm.

Prediction of customers, consumption behaviors can be used to determine the need of the materials per day.

Most of the conditions such as the weather, official holidays, location, traffic are considered in algorithm to estimate the number of possible customers for the day.

Therefore, the materials that need to be used for the day can be adjusted accordingly.

Thank You For Listening